

OBJECTIVES

- Introduce students to deploy services/docker containers into Kubernetes

GROUP DISCUSSION

Complete the following table :

Real World	Kubernetes World
Person	...
Shared apartment	...
Building with many floors	...
Real estate company ensuring apartments exist	...
City traffic system	...
City auto-builds apartments when crowded	...
City government	...

- What does it mean the word “Ephemeral” ?

INDIVIDUAL EXERCISES

Task / Exercise C1 [30 Minutes] (Deploy & Config) :

Students should consult to this tutorial for this exercise

https://drive.google.com/file/d/19vTZPIGelDH1_k1zDaLKRtdDgks1sUSZ/view?usp=classroom_web&authuser=0 : They need to ensure the following tasks are completed:

Part 1 : Start the local cluster :

- `minikube start`
- `kubectl get nodes`

Part 2 : Deploy a simple web service

- `kubectl apply -f web-deployment.yaml`
- `kubectl apply -f web-service.yaml`
- `minikube service web`
- `kubectl get pods`
- `kubectl describe pod <name>`

Task / Exercise C2 [25 Minutes] (Scaling & Assessing the load using K6)

Part 1: Manual Scaling

Students need to issue the following command to scale up:

- `kubectl scale deploy/web --replicas=5`

Later watch the pods:

- `kubectl get pods -w`

In order to scale down:

- `kubectl scale deploy/web --replicas=1`

Part 2 : Adding CPU/Limits for HPA

Edit the deployment to add resources with the file : `web-deployment.yaml`, As an Example:

```
containers:
- name: web
  image: nginx
  ports:
  - containerPort: 80
resources:
```

```
requests:
  cpu: "100m"
  memory: "64Mi"
limits:
  cpu: "500m"
  memory: "128Mi"
```

Apply:

- `kubectl apply -f web-deployment.yaml`
- `kubectl rollout status deploy/web`

Part 3 : Enabling Auto Scaling

1. Ensure metrics are available (common Minikube step):
 - `minikube addons enable metrics-server`
 - `kubectl get pods -n kube-system | grep metrics`
2. Create an HPA:
 - `kubectl autoscale deploy web --cpu-percent=50 --min=1 --max=10`
3. Observe:
 - `kubectl get hpa -w`
 - `kubectl get pods -w`

Part 4 : Testing the Auto-Scaling Mechanism via K6

1. Get the service URL:

- `minikube service web --url`

Assume it prints something like **http://YOUR_IP:XXXXX**

2. Create [load.js](#):

```
import http from 'k6/http';
import { sleep } from 'k6';
export const options = {
  stages: [
    { duration: '30s', target: 10 },
    { duration: '60s', target: 50 },
    { duration: '60s', target: 100 },
    { duration: '30s', target: 0 },
  ],
};
export default function () {
  http.get(__ENV.TARGET);
  sleep(0.1);
}
```

3. Run the following command: **Make sure to your own IP:PORT**

- `docker run --rm -i --network host grafana/k6 run -e TARGET=http://192.168.49.2:30007 - < load.js`

4. In another terminal, watch scaling:

- `kubectl get hpa -w`
- `kubectl get pods -w`

```
^C(base) imed@imed-Inspiron:~/Dropbox/Workspace/Teaching/ENSIA/EnterpriseComputing/new/Lab3$ kubectl get pods -w
NAME                                READY   STATUS              RESTARTS   AGE
web-7dd6ff9997-7gwtk                0/1     ContainerCreating   0           2s
web-7dd6ff9997-v5thx                0/1     ContainerCreating   0           2s
web-7dd6ff9997-zkm96                1/1     Running             0           12m
web-7dd6ff9997-7gwtk                1/1     Running             0           3s
web-7dd6ff9997-v5thx                1/1     Running             0           5s
^C(base) imed@imed-Inspiron:~/Dropbox/Workspace/Teaching/ENSIA/EnterpriseComputing/new/Lab3$ kubectl get hpa -w
NAME    REFERENCE          TARGETS      MINPODS   MAXPODS   REPLICAS   AGE
web     Deployment/web     cpu: 114%/50% 1          10         1           11m
web     Deployment/web     cpu: 114%/50% 1          10         3           12m
```

Task / Exercise C3 [20 Minutes] (Persistence Storage)

Part 1 :

As shown in the tutorial , Create :

- Persistence storage volume that uses Postgres Server.
- Persistence storage claim
- Service to access the Persistence storage volume.

Part 2 :

- Integrate the Document (only) Spring Boot service inside a Pod (without S3 storage)
- Configure the Document service to use persistent storage.
- Add an HPA for the document service.